

EcoPassEU: A Hybrid Blockchain Architecture for ESPR-Compliant Digital Product Passports

Fuyu Cao¹, Luxiao Cao¹, Detai Dong¹, Akshansh Rajora¹, and Tianwei Yu¹

University of Bristol, Bristol, UK

{1d25515, ne25392, mz25325, tk25240, np25024}@bristol.ac.uk

1 Introduction

Billions of products enter the European market with sustainability labels every year, and consumers have little reliable means of verification for the information included, such as “low carbon” or “ethically sourced” [29]. A study commissioned by European Commission found more than half green environmental declarations without verifiable evidence to confirm them, eroding public confidence in environmental commitments [29].

To address the credibility gap, the Ecodesign for Sustainable Products Regulation (ESPR) now mandates Digital Product Passports (DPPs) -standardised records of a products’ lifecycle- for most EU products [9]. With the implementation of compliance in 2027, reliable DPP solutions have become a market urgent need.

Our product EcoPassEU is a blockchain-based DPP platform which allows manufacturers to record product information as permanent, publicly verifiable entries on the Ethereum blockchain. Unlike traditional databases, the immutability of blockchain records means that once the data is released, it cannot be altered retrospectively, eliminating the possibility of post-hoc falsification [7]. Any user can obtain a complete and encrypted product history by entering the product ID or scanning the QR code.

EcoPassEU serves four key stakeholders. It provides SME manufacturers with a cost-effective ESPR compliance channel through a web for batch data entry and automated QR code generation. At the same time, consumers, regulators and retailers can also get open and wallet-free access. Just search for the product ID to immediately obtain the blockchain-verified product passport and supply chain map, enabling the independent verification of environmental claims without prior technical expertise.

The value of EcoPassEU is threefold. It gives manufacturers a practical pathway to meet ESPR deadlines, reducing the risk of market exclusion. The platform also combats greenwashing: securing product records via cryptographic hashing replaces brand self-reporting with independent, third-party verification. Finally, it supports the circular economy by equipping recyclers with precise material data necessary to recover resources at end-of-life.

2 Background

2.1 Digital Product Passports and the Regulatory Context

A Digital Product Passport (DPP) is a structured digital record encompassing lifecycle data—such as material composition, provenance, and end-of-life guidance—designed to inform stakeholders and support more sustainable design choices, efficient resource use, and the transition towards circular business models [1,3,6,19].

The regulatory foundation for DPPs comes from the European Green Deal and Circular Economy Action Plan [15,16]. Building on these two frameworks, the Ecodesign for Sustainable Products Regulation (ESPR) has come into force from 18 July 2024 for products in the scope of the regulation, mandating DPPs for almost all physical goods sold in the EU, regardless of where the goods are sold [9].

2.2 The Problem: Trust in Sustainability Claims

Despite regulatory momentum, most DPP implementations rely on centralised, proprietary databases [27]. This architecture encourages selective concealment of data, hindering aftermarket participants such as independent repairers [10]. Managing large volumes of lifecycle data centrally also raises important concerns regarding its storage efficiency, data integrity, and vulnerability to breaches [33]. As such, downstream actors, including importers, recyclers and end consumers, are left with no reliable means to independently verify product information.

Greenwashing compounds this problem. Broadly understood as false, vague, or unsubstantiated environmental claims [22], it is widespread: a 2020–2021 EU sweep of 344 online claims found that nearly half were potentially deceptive—often substituting verifiable details with vague terms like “conscious”—and 59% lacked accessible evidence [29]. A separate EU website screening found that 42% of sustainability claims were misleading enough to constitute improper commercial behaviour [29].

Blockchain technology offers a structural solution to this issue. Because records are replicated across a distributed network, no single party can tamper with the data without detection [7,25]. It is important to note that while blockchain cannot guarantee that the data is initially physically true (the “oracle problem”) [11], it can ensure strict tamper-evidence, meaning that upon the commitment of a manufacturer to a sustainability claim, they are held publicly accountable for that claim [35].

2.3 Related Work and Existing Platforms

Blockchain-based product traceability uses its immutability and decentralised verification to improve supply chain transparency [7]. Blockchain-based product traceability emerged as a research topic around 2015–2016, with early work

concentrated in food and agriculture before spreading to pharmaceuticals, electronics, and textiles [12,30]. More recently, research has shifted from feasibility towards scale and efficiency: for instance, Walmart’s use of IBM Food Trust — built on Hyperledger Fabric — to reduce food-item tracing time from six days to under three seconds [12] .

In the commercial field, Circularise is among the closest comparable platforms, offering DPPs with selective disclosure of sustainability data across supply chains, with applications spanning the automotive, battery, and textile sectors [8]. VeChain similarly deploys blockchain and smart contracts on its VeChainThor platform to achieve supply chain transparency, though it operates on a permissioned network with a restricted set of authority-approved validators [32]. Both platforms share a common limitation: reliance on proprietary infrastructure or permissioned networks introduces vendor dependency and reduces interoperability [4,34].

EcoPassEU takes a different approach. Rather than relying on a proprietary database, it is built on the public Ethereum blockchain and IPFS, with all rules encoded in open smart contracts. Product data is stored on IPFS, whilst only a small set of cryptographic identifiers is written on-chain, keeping costs low. The batch registration further reduces the cost of individual products and makes the platform more attractive to SMEs. Moreover, as an open source project, it does not have the risk of vendor lock-in.

2.4 Business Viability

EcoPassEU’s revenue model rests on two foundations. One is public funding: the EU’s CIRPASS-2 Project (2024—2027) explicitly subsidise DPP deployment across textiles, electronics, and construction, with dedicated support for SMEs adopting “DPP-as-a-Service” models [17]. As an open source platform, EcoPassEU is fully capable of benefitting from such grants.

The other is subscription revenue. ESPR compliance is legally mandatory for any business selling physical goods in the EU, forming a captive market independent of voluntary commitments [9]. The green technology and sustainability market is projected to grow from USD 25.47billion in 2025 to USD 73.90 billion in 2030 [24]. EcoPassEU charges a graded subscription fee based on passport volume, with optional premium tiers for API integration and compliance reporting. The platform is built on public Ethereum, supporting payment with ETH or other ERC-20 tokens, providing international users a lower-friction settlement option and removing the need for traditional banking intermediaries.

3 Design Documentation

3.1 Design Philosophy and Architectural Choices

EcoPassEU is a digital product passport (DPP) designed to provide an ESPR-compliant digital product passport and contains independently verifiable sustainability data. The design follows three design principles, each of which is aimed at specific product needs.

First, the hybrid storage model is adopted: the complete product data is fixed on IPFS through Pinata [5], and only the content identifier (CID), keccak256 integrity hash, issuer address and timestamp [35] are recorded on the chain, thus reducing gas costs. Second, the system is privacy-aware and future-proof. While full data is currently stored off-chain on public network IPFS, this architecture sets the necessary foundation for data privacy. In future versions, manufacturers will also be able to easily encrypt commercially sensitive product data before uploading it. The on-chain hash will continue to function as an integrity anchor, proving the data has not been altered without exposing the full contents to the public ledger. Third, all read queries use public RPC endpoints, allowing consumers and regulators to view any passport without holding wallets or paying gas fees.

3.2 System Architecture

The user layer comprises 3 separate actor types: manufacturers, who connect to the platform via MetaMask and sign/submit transactions, and consumers/regulators, who can query product records without holding a wallet. This is a non-exhaustive overview, as both actors must pass through the application layer — a React/Vite frontend deployed on the cloud platform Vercel that makes use of Ethers.js v6 for contract calls — to perform document uploads to IPFS, contract calls to the Sepolia network, and local integrity checks. These actor types, with differing requirements, are served by EcoPassEU (See Figure 1).

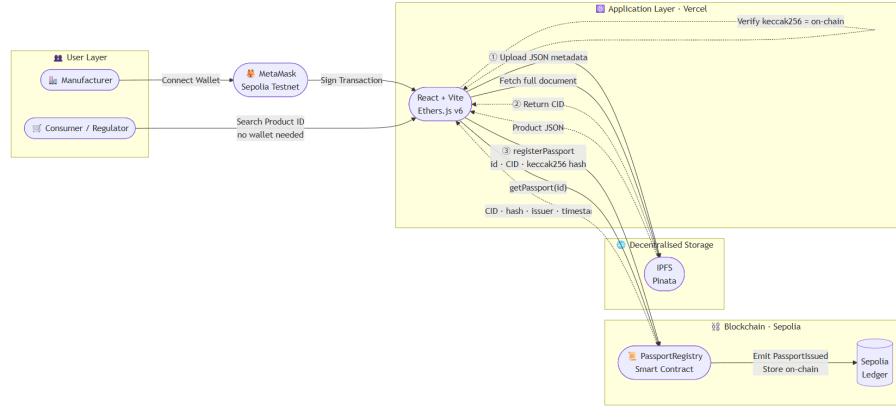


Fig. 1. Overall system architecture.

3.3 Use Cases and Workflow

EcoPassEU serves three actor types with differing technical needs, as illustrated in Figure 2.

Wallet connection via MetaMask on the Sepolia testnet (UC-01) is a common premise for all manufacturers and administrators, as each on-chain write must be signed by a wallet key pair and cryptographically linked to an accountable issuer [20,21]. To register a product, the manufacturer fills in a structured registration form (UC-02), including product details, material composition, and supply chain sources. Upon submission, the frontend serialises the form data to JSON and uploads it to IPFS via Pinata (UC-03) to receive a CID. Upon submission, the frontend serialises the form data to JSON and uploads it to IPFS via Pinata (UC-03) to receive a CID.

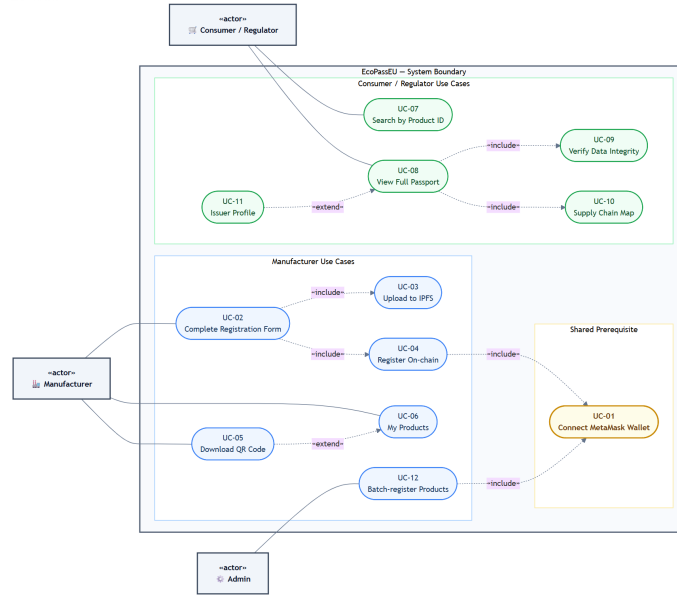


Fig. 2. EcoPassEU use case diagram.

Next, the system calculates the `keccak256` hash of the JSON locally and passes it to the `registerPassport()` function on the smart contract alongside the CID and product ID (UC-04). After the user confirms the transaction in MetaMask, a `PassportIssued` event is emitted, and a QR code with an Ether-scan link is provided for users to download (UC-05), which also accesses the product list view (UC-06). The admin role requires the same wallet connection but adds batch registration capabilities (UC-12) to reduce the unit gas cost for high-volume producers.

At the read end, consumers and regulators can verify the product without a wallet. The user submits the product ID (UC-07), and the frontend queries the contract through the public RPC endpoint and jumps to the complete product passport page (UC-08). If the product passport exists, the system will obtain the complete JSON data from IPFS and recalculate its hash value on the client: if it matches, the green integrity badge will be displayed; if it does not match, a warning (UC-09) will be displayed. The product passport page will also display

an interactive supply chain map (UC-10) and an optional issuer profile (UC-11). This separation between the write-gated and read-open flows supports the design goal of providing transparent information without setting financial or technical barriers to end users.

3.4 Sequence and State Diagrams

Figure 3 outlines the technical details behind the core application flows, focussing on the strict execution order required for successful registration. Specifically, the frontend must first obtain a valid CID from IPFS before prompting the wallet signature and triggering the smart contract. This ensures that the on-chain pointer never leads to data loss. Meanwhile, the verification flow demonstrates how the system relies entirely on the client logic to compare the acquired IPFS documents against the on-chain hash, ensuring trustless validation.

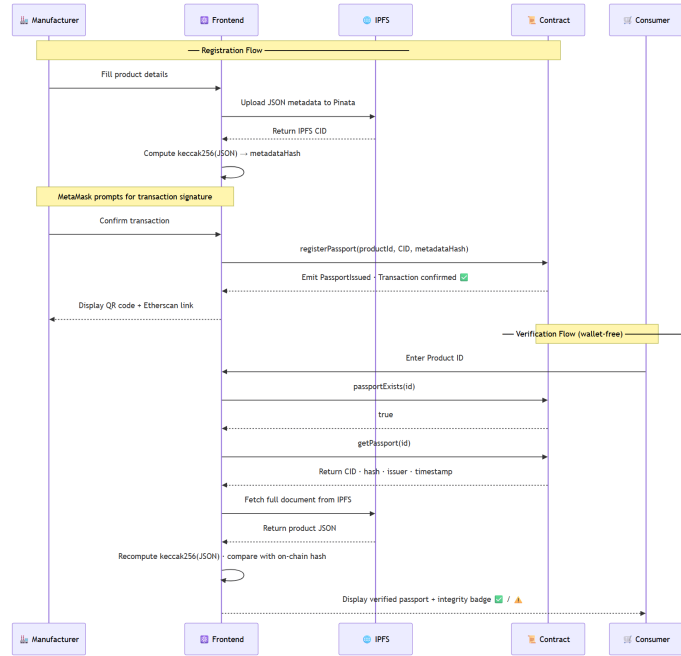


Fig. 3. UML sequence diagram.

Figure 4 illustrates a state machine diagram that describes the lifecycle that a digital passport undergoes. At the beginning of the process, physical products were always entered as **Unregistered**. The status of the digital document only changes to **IPFSStored** after the data is successfully uploaded to Pinata. Once the user signs the document, the status of the document changes to **TxPending**. After the block is mined on the Sepolia network, the digital document becomes

immutable and is then marked as **Registered**. During the final verification step, the client hash comparison will mark the record as either **Verified** when the hash is similar or **Compromised** when the data has been altered.

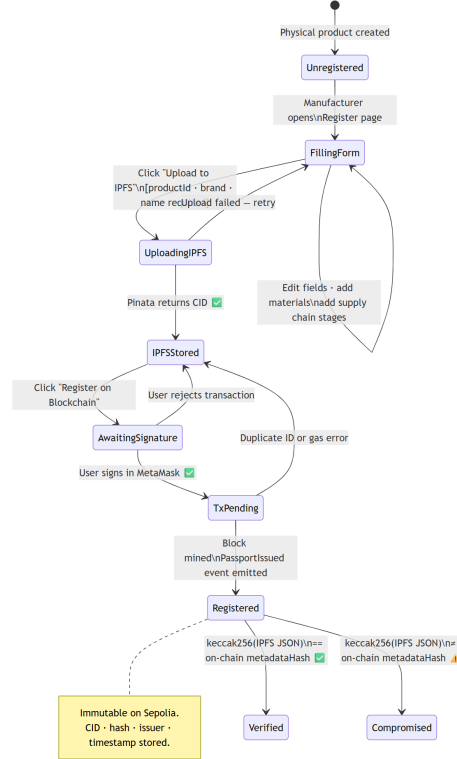


Fig. 4. UML state machine diagram.

3.5 User Interface Design

To reflect the environmental theme, the search page adopts a dark green colour scheme (#2d6a4f, #1b4332). Figure 5 shows that the search page is laid out with simple and consistent spacing. It presents a user-focused interface with a clear flow. Users can perform a search for a product by providing a product ID in the search box in the centre of the page or click on one of the pre-defined quick search buttons. No wallet connection is required. The interface begins with an empty state and upon a search is then replaced by a verified product result card. The card contains the crucial information on-chain data, such as the product issuer address, timestamp of registration on-chain, and the hash of product metadata. This allows users to instantly verify the authenticity of the product before accessing the full product passport.

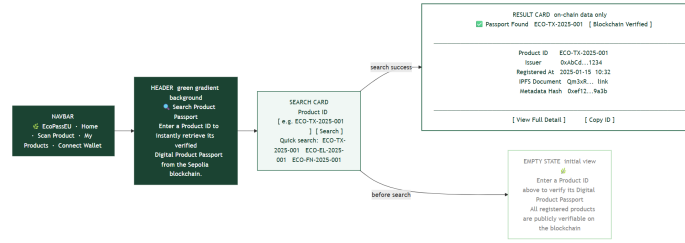


Fig. 5. Wireframe of the search interface.

Figure 6 displays the registration layout. To provide a clear representation of the user's current state in the registration process, a simple tracker is included on the page with different steps: Fill Details → Upload IPFS → Register Chain → Done. There is no hidden section; all fields within the product detail, materials, and supply chain forms are presented on one scrollable page, which avoids confusion where the user does not know which section has not yet been completed or submitted. If the page is reached without having MetaMask connected, an amber coloured banner will appear under the details section. By doing this, we can stop users from uploading their data if they have not logged in yet.

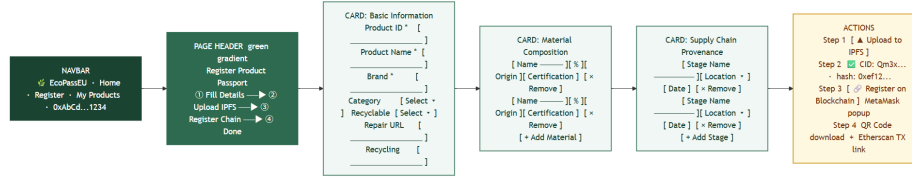


Fig. 6. Wireframe layout of the manufacturer registration form.

The result page is described in Figure 7, clearly differentiating information. The *Blockchain Record* card displays on-chain data (issuer, time, and hash), while other cards display IPFS details. It is easy to see whether the information is valid by only considering the integrity badge. It will be presented as a green checkmark if the check can be passed, or as an amber triangle if an error occurs. If a user does not install MetaMask, the page will automatically use public RPC to load information.

We've implemented a CSS Grid and a Flexbox to ensure our site works perfectly on any device. At 600px, forms will switch to a single-column layout to adapt to the mobile device screen. We had to use a 768px breakpoint for the menu and interactive map size, to make it easier for users on a tablet.

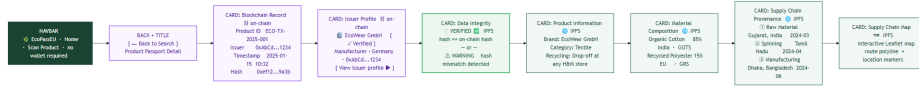


Fig. 7. Wireframe detailing the structure of the product verification page.

4 Project Execution

4.1 Core Code Analysis and Data Structures

EcoPassEU uses a hybrid architecture, separating on-chain and off-chain data to balance transparency with efficiency. The core data structure is divided across these two environments.

Off-chain, the complete lifecycle data—including category, brand, material composition, and supply chain provenance—is structured as JSON objects and pinned to IPFS via Pinata. On-chain, the `PassportRegistry.sol` smart contract uses a strictly typed `Passport` struct to store only the essential, immutable references:

Listing 1.1. Core Data Structure in `PassportRegistry.sol`

```
struct Passport {
    string ipfsCID;           // off-chain
                             document pointer
    bytes32 metadataHash;    // keccak256
                             of JSON for tamper-evidence
    address issuer;
    uint256 timestamp;
    bool exists;
}

// productId => Passport
mapping(string => Passport) private passports
;
```

This private mapping directly associates a unique product ID with its corresponding passport data for $O(1)$ retrieval. Initially, we considered using ERC-1155 tokens to represent these passports; however, we opted for this lightweight registry approach. Storing only the IPFS CID, a cryptographic hash, the issuer address, and a block timestamp on-chain significantly reduces deployment complexity and gas costs per registration, whilst keeping the full product lifecycle data off-chain on IPFS.

Listing 1.2 outlines the core registration functions and reflects a number of optimisations for Solidity:

Listing 1.2. `registerPassport` function

```
function registerPassport(
```

```

    string calldata productId,
    string calldata ipfsCID,
    bytes32 metadataHash
  ) external {
    require(!passports[productId].exists,
      "Passport already exists");
    passports[productId] = Passport({
      ipfsCID:      ipfsCID,
      metadataHash: metadataHash,
      issuer:       msg.sender,
      timestamp:    block.timestamp
    },
      exists:      true
    );
    emit PassportIssued(productId,
      ipfsCID, metadataHash,
      msg.sender, block.timestamp);
  }

```

Passing string parameters as `calldata` instead of `memory` avoids redundant data copying and reduces gas consumption. The `external` visibility modifier further improves efficiency by allowing direct argument access from `calldata`, compared with a `public` declaration. What is more, binding `issuer` to `msg.sender`—a secure EVM primitive—provides unforgeable cryptographic attribution for every passport without requiring supplementary access controls.

As a supplement to the registration logic, the `PassportIssued` event will be issued after each successful execution:

Listing 1.3. PassportIssued Event Definition

```

event PassportIssued(
  string indexed productId,
  string indexed ipfsCID,
  bytes32 metadataHash,
  address indexed issuer,
  uint256 timestamp
);

```

By designating `productId` and `issuer` as `indexed`, EVM stores these parameters in the Bloom filter of the transaction receipt. This mechanism allows off-chain clients to efficiently filter logs by specific product or manufacturer without scanning the computing overhead of each block, which represents best-practice model of establishing queryable audit tracking in Ethereum DApps [35].

4.2 Testing Strategy

To validate the reliability of `PassportRegistry`, a unit-testing suite was implemented using the Hardhat environment alongside the Chai assertion library [?], following established practices for smart contract validation [18]. Listing 1.4

shows the shared test fixture; a fresh contract instance is deployed via `beforeEach`, ensuring full isolation between cases.

Listing 1.4. Hardhat test fixture – shared setup

```
beforeEach(async function () {
    [owner, addr1] = await hre.ethers.
        getSigners();
    const Factory =
        await hre.ethers.getContractFactory("
            PassportRegistry");
    contract = await Factory.deploy();
});
```

The most critical tests concern `registerPassport`, seen in Listing 1.5. The first case confirms that `msg.sender` is persisted as the issuer, which is the foundation of the platform’s unforgeable attribution model. The second verifies that attempting to re-register an existing product ID triggers the `require` guard and reverts with the expected reason string, guaranteeing passport immutability.

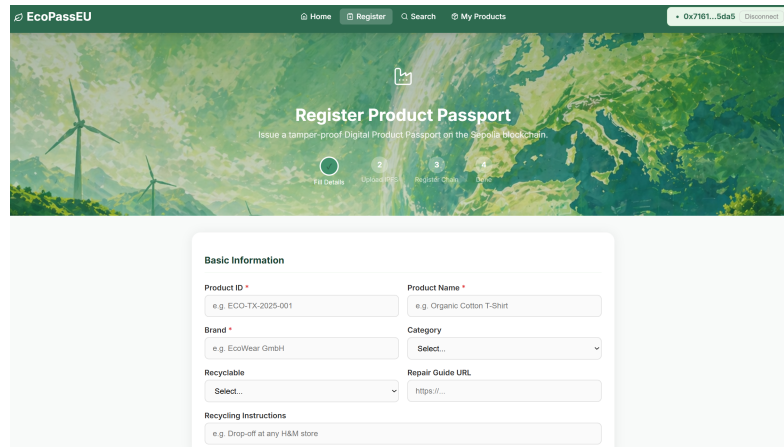
Listing 1.5. Unit tests – registerPassport: issuer storage and duplicate guard

```
it("should store the correct issuer address",
    async function () {
        await contract.registerPassport(
            PRODUCT_ID, IPFS_CID, META_HASH);
        const [, , issuer] = await contract.
            getPassport(PRODUCT_ID);
        expect(issuer).to.equal(owner.address
        );
    });

it("should revert on duplicate product ID",
    async function () {
        await contract.registerPassport(
            PRODUCT_ID, IPFS_CID, META_HASH);
        await expect(
            contract.registerPassport(PRODUCT_ID,
                IPFS_CID, META_HASH)
        ).to.be.revertedWith("Passport
            already exists");
    });
```

4.3 Final UI Showcase

Figures 8 and 9 show the deployed interface of EcoPassEU. The registration portal guides manufacturers through a structured form (with step indicator), whilst the public verification page shows the passport details, integrity status and IPFS document link (without wallet requirement).



EcoPassEU Home Register Search My Products 0x7951...5da5 Disconnected

Register Product Passport

Issue a tamper-proof Digital Product Passport on the Sepolia blockchain

1 Fill Details 2 Upload Assets 3 Register Chain 4

Basic Information

Product ID *
e.g. ECO-TX-2025-001

Product Name *
e.g. Organic Cotton T-Shirt

Brand *
e.g. EcoWear GmbH

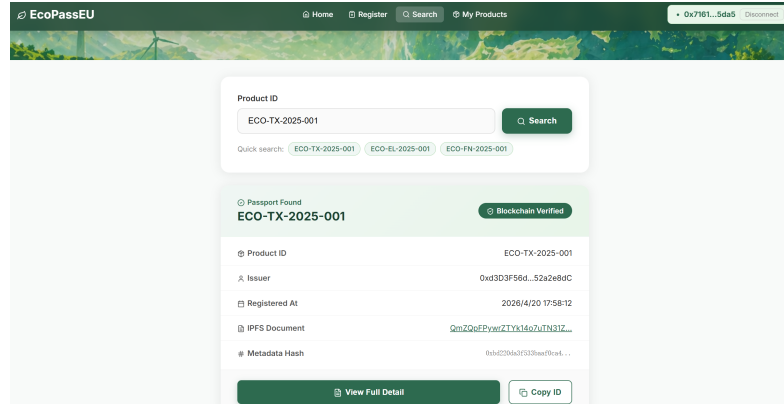
Category
Select...

Recyclable
Select...

Repair Guide URL
https://...

Recycling Instructions
e.g. Drop-off at any H&M store

Fig. 8. Deployed EcoPassEU interface: manufacturer registration portal.



EcoPassEU Home Register Search My Products 0x7951...5da5 Disconnected

Product ID
ECO-TX-2025-001 Search

Quick search: ECO-TX-2025-001 ECO-EL-2025-001 ECO-FN-2025-001

Passport Found
ECO-TX-2025-001 Blockchain Verified

Product ID	ECO-TX-2025-001
Issuer	0xd303f56d...52a2e9dC
Registered At	2026/4/20 17:58:12
IPFS Document	QmZQeFpYwZTxx16o7uIN3IZ...
Metadata Hash	8b4c255ac0753bae9a6t...

View Full Detail Copy ID

Fig. 9. Deployed EcoPassEU interface: public product verification page.

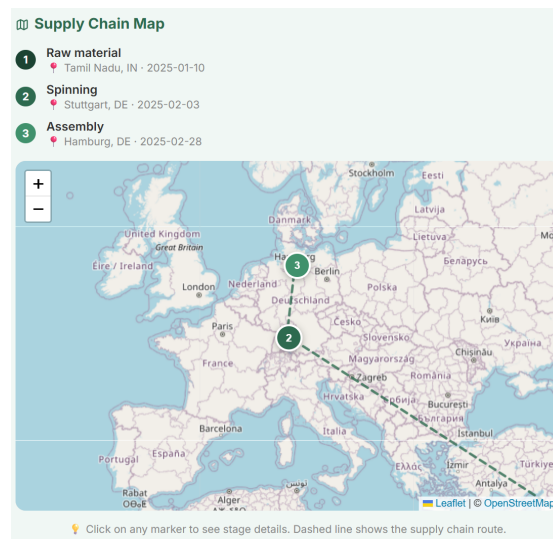


Fig. 10. Interactive supply chain map rendered on the public product verification page (UC-10).

Figure 10 illustrates the interactive supply chain map that displays every registered stage of the supply chain from a geographic perspective. By doing this, we enable consumers and regulators to follow the product’s provenance, from the sourcing of raw materials to its final assembly.

4.4 User Manual

The application is designed to be intuitive for two distinct user groups.

For Manufacturers (Registration):

1. Connect your MetaMask wallet to the Sepolia testnet using the top navigation bar.
2. Navigate to the ‘Register’ page and complete the product detail forms, including material and supply chain data.
3. Click ‘Upload to IPFS’. The system will generate a unique CID.
4. Click ‘Register on Blockchain’ and approve the transaction in the MetaMask pop-up.
5. Once confirmed, download the generated QR code to attach to your physical products.

For Consumers and Regulators (Verification):

1. No wallet connection needed. Open the EcoPassEU homepage.
2. Enter the Product ID into the central search bar, or scan the product’s QR code.
3. Review the product’s lifecycle data. There will be a green ‘Blockchain Verified’ badge, indicating that the IPFS data matches the immutable blockchain record.

5 Critical Evaluation

A per-transaction cost analysis of this DApp’s operation reveals highly optimised gas efficiency. The transaction cost can be rigorously decomposed based on standard EVM opcode pricing [35]. Each registration transaction has a base cost of 21,000 gas, with an additional cost of approximately 20,000 gas per `SSTORE` opcode for each new storage slot written. By storing heavy string data (e.g., repair guides) off-chain on IPFS, the on-chain state changes can be kept extremely lightweight. Moreover, it’s important to note that the `Passport` struct can write 5 fields, however, using `bytes32` type for `metadataHash` instead of a hex string halves the storage cost for this one field, since it fits perfectly into one 32-byte EVM storage word. The testnet deployment reveals that conducting a single registration transaction would use approximately 125,430 gas. However, it is critical to mention that there can be very high operational scalability provided by the `batchRegisterPassports` function. Since the contract iterates an array of products in one transaction, it effectively amortises the fixed 21,000-gas base cost across multiple registration transactions. Based on the test data, it has been

shown that registering 10 products in bulk reduces the average per-product cost to roughly 92,100 gas. This in turn explains the 26% per-unit cost reduction, and gives the platform potential to be economically viable for SMEs.

In terms of security, the implementation has successfully reduced the risk of tampering and unauthorised modification, which is a central concern in Ethereum smart contract security [23,31]. In smart contracts, the explicit `require` check provides a strict tamperability detection guarantee by making the registered passport immutable. In addition, the stored metadata hash establishes a cryptographic commitment between the on-chain registry and the off-chain IPFS load. If the malicious actor tries to intercept and tamper with the IPFS document, the cryptographic hash of the tampered document will no longer match the immutable record saved on the chain, thus immediately invalidating the passport. Unlike existing solutions that rely on permissioned nodes or proprietary databases, EcoPassEU achieves enterprise-level privacy protection on a fully open network through the IPFS hash pairing mechanism.

The contract structurally avoids several common loopholes in Ethereum. The re-entry attack [23] is not applicable because the contract neither accepts nor transfers Ether, eliminating the external call path used by reentrancy attack. Integer overflow has been mitigated by Solidity 0.8.x’s built-in checked arithmetic, which reverts automatically on overflow without requiring the SafeMath library [14]. The `private` visibility of `passports` mapping prevents direct external enumeration; all reading are controlled by the typed `getPassport()` and `passportExists()` view functions, which are declared as `view`, therefore cost zero gas when called off-chain.

Although the proposed EcoPassEU system presents strong advantages, it does have some limitations that arise from the decentralised architecture. While the on-chain verification process is extremely fast, obtaining the complete JSON from public IPFS gateways can have some longer latency compared to centralised servers. First and most importantly, even though the batch registration function is very affordable on a testnet, executing such loop-based operations directly on the Ethereum mainnet will get immensely pricey during congested networks, so the production version of EcoPassEU is intended primarily for deployment on Layer-2 networks (Polygon, for example, or Arbitrum). This approach will still allow EVM compatibility and security while also making near-zero transaction fees for long-term financial viability of SMEs [28].

6 Conclusion

In this work, we show that a proof-of-concept implementation (EcoPassEU) builds a workably tamper-proof infrastructure for Digital Product Passports according to the ESPR. By integrating Ethereum smart contracts [2] with the off-chain system, IPFS, we create independently verifiable sustainability records that do not compel end users to handle crypto wallets. Further, a combination of a `keccak256` hash pairing system [26] with Ethereum’s Proof-of-Stake consensus [13] solves the fundamental integrity problem without the high carbon footprint

that typically accompanies blockchains. Finally, the feature of batch registration results in a 26% reduction in gas costs, bringing much greater access for small and medium-sized enterprises (SMEs).

Nonetheless, the prototype has several limitations in practical use that must be overcome before launch to the public. First, using the testnet exclusively to test on the Sepolia testnet means that gas costs are untested in practice. Specifically, this is true for the loop-based `batchRegisterPassports` function, which in mainnet congestion could become prohibitively expensive. Another crucial problem lies in the off-chain storage, where, at the moment, it is purely dependent on Pinata. In the case of this single gateway being unavailable, the documents it links to will become inaccessible, leaving the passport cryptographically correct, but useless in use. As well, the whole system is completely open; there are no identity checks, allowing any wallet address to issue a passport, so providing no confidence that it came from the genuine manufacturer. Finally and most crucially, the strict immutability of the blockchain does not allow users to fix mistakes they make in good faith or to withdraw invalidated entries. This is a large issue for regulators, as, at present, there is also no upgradeable proxy pattern in the current smart contract design.

EcoPassEU will need a clear roadmap to transition from a proof-of-concept into a production-ready environment. As outlined earlier, the platform needs to be migrated into a Layer-2 solution to ensure costs are affordable when scaled. Additionally, the current proof-of-concept is not set up to have a formal identity verification mechanism. In future versions, W3C Decentralised Identifiers (DIDs) and Verifiable Credentials [36,37] should be included in the system. Not only will this link a manufacturer’s wallet to a legally verified entity, but it solves the “oracle problem” without sacrificing corporate privacy.

Finally, addressing the strict immutability of the blockchain, an upgradeable and controlled proxy pattern should be considered. This would give regulators and manufacturers the ability to amend honest errors or update compliance requirements, without compromising historical passport records.

References

1. Adisorn, T., Tholen, L., Götz, T.: Towards a digital product passport fit for contributing to a circular economy. *Energies* **14**, 2289 (2021). <https://doi.org/10.3390/en14082289>
2. Antonopoulos, A.M., Wood, G.: *Mastering Ethereum: Building Smart Contracts and DApps*. O'Reilly Media (2018)
3. Basal, M., Demircioglu, A.: Digital product passport in marketing and the future of sustainable development. *American Journal of Industrial and Business Management* (2024). <https://doi.org/10.4236/ajibm.2024.145039>
4. Belchior, R., Vasconcelos, A., Guerreiro, S., Correia, M.: A survey on blockchain interoperability: Past, present, and future trends. *ACM Computing Surveys* **54**, 1–41 (2020). <https://doi.org/10.1145/3471140>
5. Benet, J.: IPFS – content addressed, versioned, P2P file system (2014), <https://arxiv.org/abs/1407.3561>
6. van Capelleveen, G., Vegter, D., Olthaar, M., van Hillegersberg, J.: The anatomy of a passport for the circular economy: a conceptual definition, vision and structured literature review. *Resources, Conservation & Recycling Advances* (2023). <https://doi.org/10.1016/j.rcradv.2023.200131>
7. Casino, F., Dasaklis, T.K., Patsakis, C.: A systematic literature review of blockchain-based applications: Current status, classification and open issues. *Telematics and Informatics* **36**, 55–81 (2019). <https://doi.org/10.1016/j.tele.2018.11.006>
8. Circularise: Circularise: Product traceability platform for supply chain compliance. <https://www.circularise.com/> (2025), accessed: April 2026
9. Circularise: A guide to the ecodesign for sustainable products regulation (ESPR). <https://www.circularise.com/blogs/a-guide-to-the-ecodesign-for-sustainable-products-regulation-espr> (2025), accessed: April 2026
10. Ducuing, C., Reich, R.: Data governance: Digital product passports as a case study. *Competition and Regulation in Network Industries* **24**, 3–23 (2023). <https://doi.org/10.1177/17835917231152799>
11. Egberts, A.: *The Oracle Problem — An Analysis of how Blockchain Oracles Undermine the Advantages of Decentralized Ledger Systems*. Master's thesis, EBS Universität für Wirtschaft und Recht (December 2017). <https://doi.org/10.2139/ssrn.3382343>, available at SSRN: <https://ssrn.com/abstract=3382343>
12. Ellahi, R.M., Wood, L.C., Bekhit, A.: Blockchain-based frameworks for food traceability: A systematic review. *Foods* **12** (2023). <https://doi.org/10.3390/foods12163026>
13. Ethereum Foundation: The merge. <https://ethereum.org/en/roadmap/merge/> (2022), accessed: April 2026
14. Ethereum Foundation: Solidity language documentation. <https://docs.soliditylang.org/en/v0.8.20/> (2024), version 0.8.20. Accessed: April 2026
15. European Commission: The european green deal. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en (2019), accessed: 28 April 2026
16. European Commission: A new circular economy action plan: For a cleaner and more competitive europe. Communication from the Commission COM(2020) 98 final, European Commission, Brussels (2020)

17. European Commission: CIRPASS-2: Common infrastructure and flagship deployments for the European Digital Product Passport. <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/43152860/101158775> (2024), project ID: 101158775. Accessed: April 2026
18. Ferreira, J.F., Cruz, P., Durieux, T., Abreu, R.: Smartbugs: a framework to analyze solidity smart contracts. In: Proceedings of the 35th IEEE/ACM International Conference on Automated Software Engineering. pp. 1349–1352. ACM (2020). <https://doi.org/10.1145/3324884.3415298>
19. Jansen, M., Meisen, T., Plociennik, C., Berg, H., Pomp, A., Windholz, W.: Stop guessing in the dark: Identified requirements for digital product passport systems. *Syst.* **11**, 123 (2023). <https://doi.org/10.3390/systems11030123>
20. Jorgensen, K.P., Beck, R.: Universal wallets. *Business & Information Systems Engineering* **64**, 115–125 (2022). <https://doi.org/10.1007/s12599-021-00736-6>
21. Keršič, V., Vidovic, U., Vrecko, A., Domajnko, M., Turkanović, M.: Orchestrating digital wallets for on- and off-chain decentralized identity management. *IEEE Access* **11**, 78135–78151 (2023). <https://doi.org/10.1109/access.2023.3299047>
22. Kundi, V., Ernszt, I.: The phenomenon of greenwashing: An analysis of the hungarian regulation. *Journal of Sustainability Research* (2024). <https://doi.org/10.20900/jsr20240066>
23. Luu, L., Chu, D.H., Olickel, H., Saxena, P., Hobor, A.: Making smart contracts smarter. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security. pp. 254–269. ACM (2016). <https://doi.org/10.1145/2976749.2978309>
24. MarketsandMarkets: Green technology & sustainability market worth \$73.90 billion by 2030. <https://www.marketsandmarkets.com/PressReleases/green-technology-and-sustainability.asp> (2025), accessed: April 2026
25. Nakamoto, S.: Bitcoin: A peer-to-peer electronic cash system. <https://bitcoin.org/bitcoin.pdf> (2008), accessed: April 2026
26. National Institute of Standards and Technology: SHA-3 standard: Permutation-based hash and extendable-output functions. Tech. Rep. FIPS PUB 202, NIST (2015). <https://doi.org/10.6028/NIST.FIPS.202>
27. Psarommatis, F., May, G.: Digital product passport: A pathway to circularity and sustainability in modern manufacturing. *Sustainability* (2024). <https://doi.org/10.3390/su16010396>
28. Radomski, W., Cooke, A., Castonguay, P., Therien, J., Binet, E., Sandford, R.: Erc-1155: Multi token standard. <https://eips.ethereum.org/EIPS/eip-1155> (2018), ethereum Improvement Proposal, created 17 June 2018
29. Silva, D.: The fight against greenwashing in the european union. *UNIO – EU Law Journal* (2021). <https://doi.org/10.21814/unio.7.2.4029>
30. Tian, F.: An agri-food supply chain traceability system for China based on RFID & blockchain technology. In: 2016 13th International Conference on Service Systems and Service Management (ICSSSM). IEEE (2016). <https://doi.org/10.1109/ICSSSM.2016.7538424>
31. Tsankov, P., Dan, A., Drachsler-Cohen, D., Gervais, A., Bünzli, F., Vechev, M.: Securify: Practical security analysis of smart contracts. In: Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security. pp. 67–82. ACM (2018). <https://doi.org/10.1145/3243734.3243780>
32. VeChain: Vechain official: Real-world impact blockchain technology. <https://vechain.org/> (2026), accessed: April 2026

33. Voulgaridis, K., Lagkas, T.D., Angelopoulos, C., Boulogeorgos, A.A.A., Argyriou, V., Sarigiannidis, P.G.: Digital product passports as enablers of digital circular economy: a framework based on technological perspective. *Telecommunication Systems* **85**, 699–715 (2024). <https://doi.org/10.1007/s11235-024-01104-x>
34. Wang, G., Wang, Q., Chen, S.: Exploring blockchains interoperability: A systematic survey. *ACM Computing Surveys* **55**, 1–38 (2023). <https://doi.org/10.1145/3582882>
35. Wood, G.: Ethereum: A secure decentralised generalised transaction ledger. Tech. rep., Ethereum Foundation (2014), yellow Paper. Available at <https://ethereum.github.io/yellowpaper/paper.pdf>
36. World Wide Web Consortium: Decentralized identifiers (DIDs) v1.1. <https://www.w3.org/TR/did-1.1/> (2025), w3C Recommendation. Accessed: April 2026
37. World Wide Web Consortium: Verifiable credentials data model v2.0. <https://www.w3.org/TR/vc-data-model/> (2025), w3C Recommendation, 15 May 2025. Accessed: April 2026

References

1. Adisorn, T., Tholen, L., Götz, T.: Towards a digital product passport fit for contributing to a circular economy. *Energies* **14**, 2289 (2021). <https://doi.org/10.3390/en14082289>
2. Antonopoulos, A.M., Wood, G.: *Mastering Ethereum: Building Smart Contracts and DApps*. O'Reilly Media (2018)
3. Basal, M., Demircioglu, A.: Digital product passport in marketing and the future of sustainable development. *American Journal of Industrial and Business Management* (2024). <https://doi.org/10.4236/ajibm.2024.145039>
4. Belchior, R., Vasconcelos, A., Guerreiro, S., Correia, M.: A survey on blockchain interoperability: Past, present, and future trends. *ACM Computing Surveys* **54**, 1–41 (2020). <https://doi.org/10.1145/3471140>
5. Benet, J.: IPFS – content addressed, versioned, P2P file system (2014), <https://arxiv.org/abs/1407.3561>
6. van Capelleveen, G., Vegter, D., Olthaar, M., van Hillegersberg, J.: The anatomy of a passport for the circular economy: a conceptual definition, vision and structured literature review. *Resources, Conservation & Recycling Advances* (2023). <https://doi.org/10.1016/j.rcradv.2023.200131>
7. Casino, F., Dasaklis, T.K., Patsakis, C.: A systematic literature review of blockchain-based applications: Current status, classification and open issues. *Telematics and Informatics* **36**, 55–81 (2019). <https://doi.org/10.1016/j.tele.2018.11.006>
8. Circularise: Circularise: Product traceability platform for supply chain compliance. <https://www.circularise.com/> (2025), accessed: April 2026
9. Circularise: A guide to the ecodesign for sustainable products regulation (ESPR). <https://www.circularise.com/blogs/a-guide-to-the-ecodesign-for-sustainable-products-regulation-espr> (2025), accessed: April 2026
10. Ducuing, C., Reich, R.: Data governance: Digital product passports as a case study. *Competition and Regulation in Network Industries* **24**, 3–23 (2023). <https://doi.org/10.1177/17835917231152799>
11. Egberts, A.: *The Oracle Problem — An Analysis of how Blockchain Oracles Undermine the Advantages of Decentralized Ledger Systems*. Master's thesis, EBS Universität für Wirtschaft und Recht (December 2017). <https://doi.org/10.2139/ssrn.3382343>, available at SSRN: <https://ssrn.com/abstract=3382343>
12. Ellahi, R.M., Wood, L.C., Bekhit, A.: Blockchain-based frameworks for food traceability: A systematic review. *Foods* **12** (2023). <https://doi.org/10.3390/foods12163026>
13. Ethereum Foundation: The merge. <https://ethereum.org/en/roadmap/merge/> (2022), accessed: April 2026
14. Ethereum Foundation: Solidity language documentation. <https://docs.soliditylang.org/en/v0.8.20/> (2024), version 0.8.20. Accessed: April 2026
15. European Commission: The european green deal. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en (2019), accessed: 28 April 2026
16. European Commission: A new circular economy action plan: For a cleaner and more competitive europe. Communication from the Commission COM(2020) 98 final, European Commission, Brussels (2020)

17. European Commission: CIRPASS-2: Common infrastructure and flagship deployments for the European Digital Product Passport. <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/43152860/101158775> (2024), project ID: 101158775. Accessed: April 2026
18. Ferreira, J.F., Cruz, P., Durieux, T., Abreu, R.: Smartbugs: a framework to analyze solidity smart contracts. In: Proceedings of the 35th IEEE/ACM International Conference on Automated Software Engineering. pp. 1349–1352. ACM (2020). <https://doi.org/10.1145/3324884.3415298>
19. Jansen, M., Meisen, T., Plociennik, C., Berg, H., Pomp, A., Windholz, W.: Stop guessing in the dark: Identified requirements for digital product passport systems. *Syst.* **11**, 123 (2023). <https://doi.org/10.3390/systems11030123>
20. Jorgensen, K.P., Beck, R.: Universal wallets. *Business & Information Systems Engineering* **64**, 115–125 (2022). <https://doi.org/10.1007/s12599-021-00736-6>
21. Keršič, V., Vidovic, U., Vrecko, A., Domajnko, M., Turkanović, M.: Orchestrating digital wallets for on- and off-chain decentralized identity management. *IEEE Access* **11**, 78135–78151 (2023). <https://doi.org/10.1109/access.2023.3299047>
22. Kundi, V., Ernszt, I.: The phenomenon of greenwashing: An analysis of the hungarian regulation. *Journal of Sustainability Research* (2024). <https://doi.org/10.20900/jsr20240066>
23. Luu, L., Chu, D.H., Olickel, H., Saxena, P., Hobor, A.: Making smart contracts smarter. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security. pp. 254–269. ACM (2016). <https://doi.org/10.1145/2976749.2978309>
24. MarketsandMarkets: Green technology & sustainability market worth \$73.90 billion by 2030. <https://www.marketsandmarkets.com/PressReleases/green-technology-and-sustainability.asp> (2025), accessed: April 2026
25. Nakamoto, S.: Bitcoin: A peer-to-peer electronic cash system. <https://bitcoin.org/bitcoin.pdf> (2008), accessed: April 2026
26. National Institute of Standards and Technology: SHA-3 standard: Permutation-based hash and extendable-output functions. Tech. Rep. FIPS PUB 202, NIST (2015). <https://doi.org/10.6028/NIST.FIPS.202>
27. Psarommatis, F., May, G.: Digital product passport: A pathway to circularity and sustainability in modern manufacturing. *Sustainability* (2024). <https://doi.org/10.3390/su16010396>
28. Radomski, W., Cooke, A., Castonguay, P., Therien, J., Binet, E., Sandford, R.: Erc-1155: Multi token standard. <https://eips.ethereum.org/EIPS/eip-1155> (2018), ethereum Improvement Proposal, created 17 June 2018
29. Silva, D.: The fight against greenwashing in the european union. *UNIO – EU Law Journal* (2021). <https://doi.org/10.21814/unio.7.2.4029>
30. Tian, F.: An agri-food supply chain traceability system for China based on RFID & blockchain technology. In: 2016 13th International Conference on Service Systems and Service Management (ICSSSM). IEEE (2016). <https://doi.org/10.1109/ICSSSM.2016.7538424>
31. Tsankov, P., Dan, A., Drachsler-Cohen, D., Gervais, A., Bünzli, F., Vechev, M.: Securify: Practical security analysis of smart contracts. In: Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security. pp. 67–82. ACM (2018). <https://doi.org/10.1145/3243734.3243780>
32. VeChain: Vechain official: Real-world impact blockchain technology. <https://vechain.org/> (2026), accessed: April 2026

33. Voulgaridis, K., Lagkas, T.D., Angelopoulos, C., Boulogeorgos, A.A.A., Argyriou, V., Sarigiannidis, P.G.: Digital product passports as enablers of digital circular economy: a framework based on technological perspective. *Telecommunication Systems* **85**, 699–715 (2024). <https://doi.org/10.1007/s11235-024-01104-x>
34. Wang, G., Wang, Q., Chen, S.: Exploring blockchains interoperability: A systematic survey. *ACM Computing Surveys* **55**, 1–38 (2023). <https://doi.org/10.1145/3582882>
35. Wood, G.: Ethereum: A secure decentralised generalised transaction ledger. Tech. rep., Ethereum Foundation (2014), yellow Paper. Available at <https://ethereum.github.io/yellowpaper/paper.pdf>
36. World Wide Web Consortium: Decentralized identifiers (DIDs) v1.1. <https://www.w3.org/TR/did-1.1/> (2025), w3C Recommendation. Accessed: April 2026
37. World Wide Web Consortium: Verifiable credentials data model v2.0. <https://www.w3.org/TR/vc-data-model/> (2025), w3C Recommendation, 15 May 2025. Accessed: April 2026

A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Deployment

- **Smart Contract Address(es):** [Insert Sepolia testnet addresses here]
- **Etherscan Links:** [Insert links to verified contracts on Etherscan]

A.2 Project Links

- **Deployed Website:** <https://ftgp-2526-group1-11.vercel.app/home>
- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11

A.3 Transaction Hashes

- **Deployment TX:** [Insert Hash] - Explanation.

A.4 Equity Shares & Contributions

- Luxiao Cao: [Weight]
- Akshansh Rajora: [Weight]
- Fuyu Cao: [Weight]
- Tianwei Yu: [Weight]
- Detai Dong: [Weight]

[Insert Screenshot of GitHub - Insights - Contributor here]

A.5 JIRA Project Board

[Insert screenshots of 'Summary', 'timeline', and 'backlog' pages here]

A.6 Individual Reflection Essays

Luxiao Cao [Insert 500-word reflection here]

Akshansh Rajora [Insert 500-word reflection here]

Fuyu Cao During the FTGP group project, I mainly contributed to the frontend presentation, visual refinement, and part of the data preparation for our DApp, EcoPassEU. Our project focused on building a privacy-aware Digital Product Passport platform for sustainable consumer goods in Europe. Compared with some of my teammates who worked more on the smart contract logic or report writing, my role was more related to improving how the project was presented and how users would interact with it. I helped optimise the page design, adjust the layout of several pages, improve the homepage background and visual consistency, and design the project logo. In addition, I also processed part of the dataset so that it could be used more clearly in the frontend demonstration.

At the beginning of the project, I did not have a very strong background in frontend development, so some tasks seemed unfamiliar to me. For example, when I first looked at the project files, I was not very confident about how to modify the UI, replace images, or update components correctly. However, through this project I gradually became more comfortable with reading the code structure, locating the relevant files, and making direct changes to the interface. I learned how the React frontend was organised, how local changes could be previewed on the browser, and how visual elements such as logos, banners, and page sections could be integrated into the project. Although these may seem like small tasks compared with blockchain deployment, I realised that they are still very important because they directly affect the clarity, usability, and professionalism of the final product.

Another useful part of my work was helping with the dataset. The raw data was not in a form that could be directly used in the project, so I helped clean part of it and organise sample data that was easier for the frontend to display. This made me understand that in a real project, the gap between raw data and usable system data is often larger than expected. Even if the final interface looks simple, a lot of supporting work is needed behind it.

From a teamwork perspective, this project also taught me a lot. I learned that not every contribution has to be the “main technical core” in order to be valuable. In a group project, different members naturally take responsibility for different parts, and what matters is whether each person helps move the project forward. In my case, improving the UI, preparing data, and helping with presentation materials were practical contributions that supported the overall result. I also became more aware of the importance of communication, especially when multiple people are editing code, updating files, and trying to keep the project consistent.

Overall, this project was a meaningful experience for me. It not only improved my understanding of blockchain-based application development, but also gave me more confidence in contributing to a technical team project in my own way. I learned that building a complete product requires not only backend logic and smart contracts, but also design, data preparation, usability, and coordination. This experience helped me better understand how a group project comes together as a whole.

Tianwei Yu [Insert 500-word reflection here]

Detai Dong [Insert 500-word reflection here]

A.7 Team Sprint Reports

[Insert chronological sprint reports here]

A.8 Complete Source Code

Smart Contracts

// Insert complete Solidity code here

Frontend Core Files

// Insert modified/relevant frontend files (e.g., page.js) here